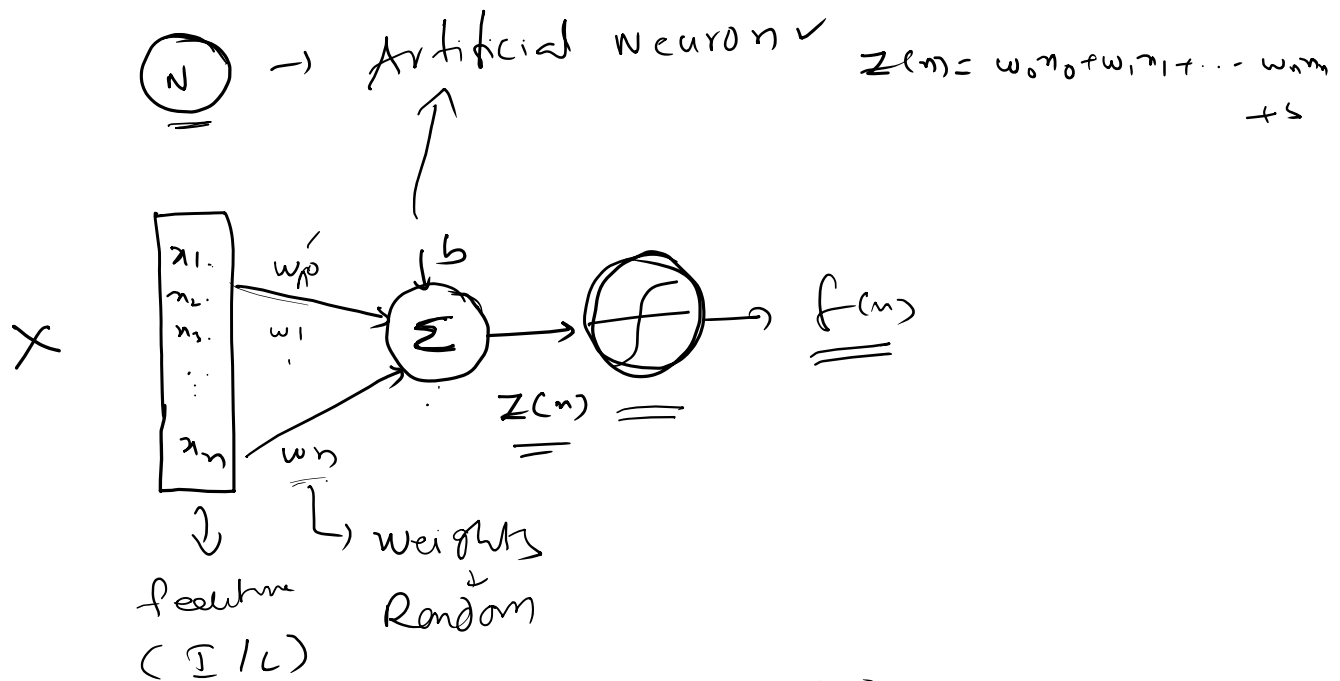




two inputs

① x_1 : hours studied

② x_2 : hours left the night before

$$z = w_1 x_1 + w_2 x_2 + b$$

$$\hat{y} = \frac{1}{1 + e^{-z}} \quad (\text{Sigmoid})$$

$\geq 0.5 \rightarrow 1$
 $\leq 0.5 \rightarrow 0$

$$w_1 = 2, \quad w_2 = 1.5, \quad b = -15$$

$$x_1 = 6, \quad x_2 = 5$$

Weighted Sum ✓

$$z = w_1(x_1) + w_2(x_2) + b$$

$$= 2(6) + (1.5)5 - 15 = 12 + 7.5 - 15 = 4.5$$

Sigmoid probability

$$\hat{y} = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-4.5}} = 0.9$$

pass ≥ 0.5 fail < 0.5 → ①

$$x_1 = 3$$

$$x_2 = 3$$

weighted sum

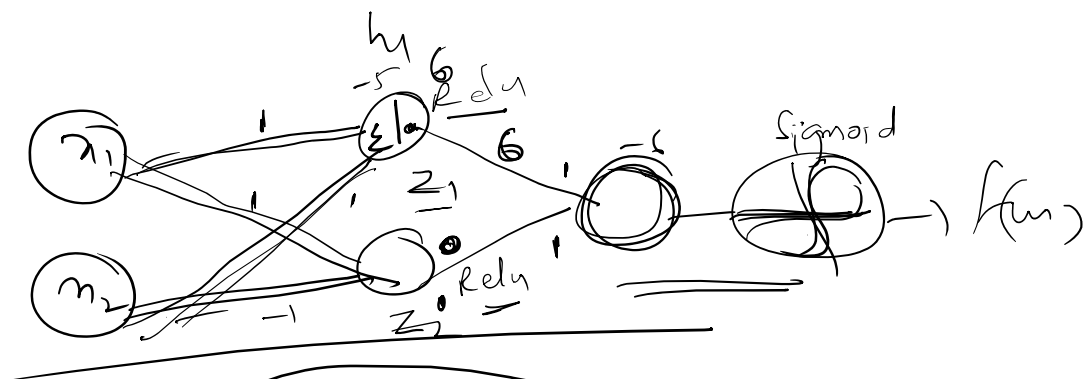
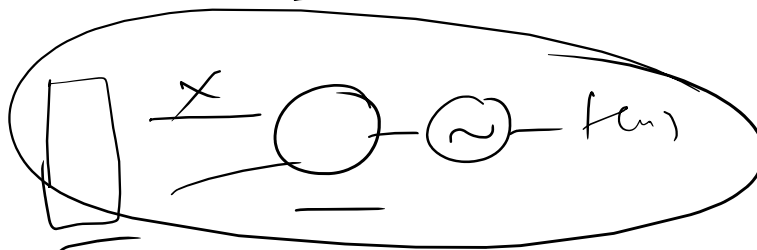
$$\begin{aligned} z &= w_1(x_1) + w_2(x_2) + b \\ &= 2(3) + 1.5(3) - 15 \\ &= 6 + 4.5 - 15 = 10.5 - 15 = -4.5 \end{aligned}$$

Sigmoid: $\frac{1}{1+e^{-z}} = \frac{1}{1+e^{-(-4.5)}} = \frac{1}{1+e^{4.5}} = \underline{\underline{0.01}}$

0.5 $\rightarrow$ fail

weights: w_1, w_2

$$\text{relu}(x) = \max(0, x)$$



$$z_1 = 1(x_1) + 1(x_2) - 5 \rightarrow \text{act} = \underline{\text{relu}}(z_1)$$

$$z_2 = 1(x_1) - 1(x_2) + 0 \rightarrow \text{act} = \underline{\text{relu}}(z_2)$$

6, 5

(0, 6) (0, -4)

$$z = 1(6) + 1(5) - 8 = 6$$

(0, 1) 0

$$z_1 = 1(6) + 1(8) - 8 = 6$$

$$z_2 = 1(6) - 1(5) - 0 = 6 - 5 = 1$$

$$z_{out} = 6(1) + 1(1) - 8 = 1 = \frac{1}{1 + e^{-z}}$$

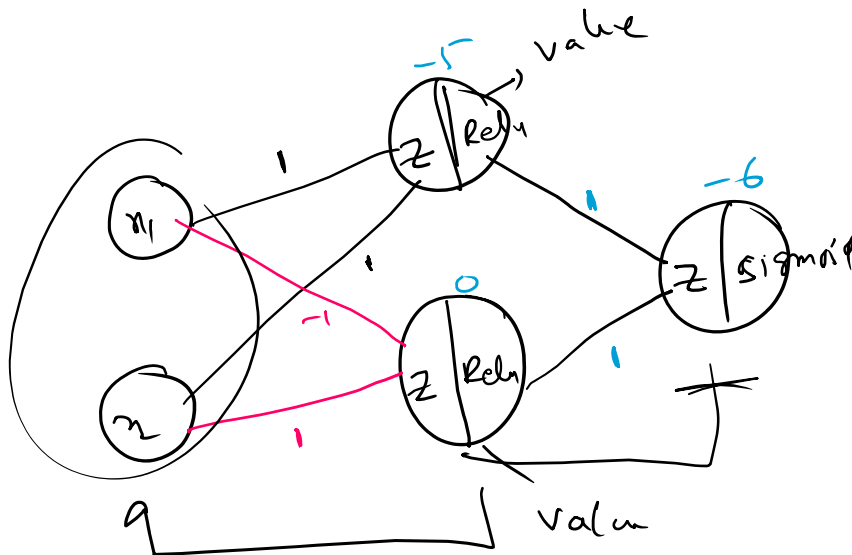
3, 3

= 0.9999999999999999

$$z_1 = 1(3) + 1(3) - 5 = 3 + 3 - 5 = 1$$

$$z_2 = 1(3) - 1(3) - 0 = 0$$

$$z_{out} = (1)(1) + 0(0) + 6 = 1 - 6 = -5 = \frac{1}{1 + e^{-(-5)}} = 0.0067$$



model = Sequential()

model.add(Dense(2))

model.add(Dense(2, input_dim=2, activation='relu'))

model.add(Dense(1, activation='sigmoid'))